RPMGLOBAL

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AMT EPF Adapter SDK

Asset Statistics 9.0



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Document History

Version	Date	Description	Author
1.0.1	24-Nov-2017	Initial Document	Monisankar Baruah & Andre Britz
1.0.1	28-Nov-2017	Added Downtime	Monisankar Baruah
1.0.2	06-Dec-2017	Updated document post review	Monisankar Baruah
1.0.3	15-Jan-2018	Correct Figure references	Dan Spathonis
	20-Feb-2018	Update the addresses	Tim McDonald
1.1.0	26-Apr-2018	Added Downtime Export	Monisankar Baruah
	25-May-2018	Review and Approve	Andre Britz
	28-May-2018	Added UTC Date fields	Andre Britz
2.0.0	5-Sep-2018	Reformat/Rename	Grant Johnson
2.0.1	9-Dec-2020	Support for EPF 2.3//2.4	Grant Johnson
2.2.5	20-Jul-2022	Added ProcessUsageProfile	Wayne Simpkins
2.2.6	27-April-2023	Clarified use of actual and planned times in Downtime Event entity for ProcessDowntimeEvent() and the response to GetDowntimeEvent() (which is ShowDowntimeEvent)	Wayne Simpkins
2.3.0	12-July-2023	Added about using AMT function 'Configure External Downtime Equipment' to map external equipment for downtime integration.	Wayne Simpkins
8.25	11-Sep-2023	Added details for ASYNC processing	Wayne Simpkins
8.26	01-Nov-2023	Add ASYNC chunk processing Added ProcessUsageReading to supported ASYNC functions. **Breaking Change:** The Response message to ProcessUsageReading has been changed. Refer to the release notes for examples of the change.	Wayne Simpkins
8.27	01-Jun-2024	Corrections to definition for UsageProfile and UsageSteps	Wayne Simpkins
8.28	01-Oct-2024	Added ASYNC support for ProcessMeasurement, ProcessDowntimeEvent and GetDowntimevent. **Breaking Changes** : These response types have been changed to match the XSD: • G**etDowntimeEventResponse** changed to **ShowDownTimeEvent** • **ProcessDownTimeResponse** changed to **RespondDowntimeEvent**.	Wayne Simpkins

Version	Date	Description	Author
		• **ProcessMeasurementResponse** changed to **RespondMeasurement**. Added a note about needing SiteName for ProcessDowntimeEvent if AMT is being used with the Downtime Import Wizard. Added support OAuth2 authentication	
8.28.2	10-Apr-2025	8.28.2 release	Wayne Simpkins
9.0	11-Jul-2025	**Breaking Change:** The Response to GetUsageReading has been corrected to match the XSD. It now correctly uses ShowUsageReading. All inbound integration functions now supported for ASYNC processing. Moved details of synchronous/asynchronous processing, and authentication options to central overview document.	Wayne Simpkins

Version Amendments

The following modifications were made to this document since v0.

Item	Description
General	Added new functions – GetDowntimeEvent and ShowDowntimeEvent.
General	Added UTC field types to all DateTime type fields. Update business rules.

1 Business Background

Usage refers to Meter readings for an asset to measure the asset's usage. The Usage Interface is designed to automate the periodic update of Meter data in AMT where the meter readings are captured in an external system. AMT also provides a facility to capture Meter data natively. Where required, these readings can be exposed allowing and external system to retrieve existing usages from AMT.

Measurements taken in external systems can be imported into AMT. These measurements can be used to report on the consumption of various items (for example, fuel, oil), and measurements can also be translated for use in the AMT CBM module according to translation rules defined by users in AMT.

Asset Downtime recorded in external systems (for example, dispatch systems) can be imported into AMT. AMT users can then take advantage of the advanced downtime reporting provided by AMT.

2 Considerations and Assumptions

2.1 Usage Readings

AMT can import Actual meter readings or Cumulative meter readings. If actual meter readings are imported, AMT users may need to maintain the offset. The offset is the difference between a usage reading and the cumulative usage reading. Often a meter may be replaced, which requires an offset to be applied. In the case of a change-out meter, the cumulative reading is the meter reading + offset.

2.2 Measurements

A record is not imported if any of the following occur:

- The model does not exist.
- The equipment does not exist.
- Measurement Code (Measurement Type Code + Element Code + Compartment) does not exist.
- Any required field is missing.
- Measurement Time is in the future.

A record is marked as "ignored" if there is no translation rule for the measurement reading. However, the record is imported.

All required referenced fields will be set up in AMT.

2.3 Downtime

External Downtime import is set up as a separate feature in AMT and is required to be enabled at an Organisational level.

There is a requirement to map External Assets and Sites to AMT Assets and Sites. AMT users can configure this in AMT using the forms provided by AMT.

An AMT installation can be configured to automatically create AMT Downtime Events when a Downtime Notification is imported from an external system.

Alternatively, an AMT installation can be configured so that an AMT user uses a feature called **Downtime Import Wizard** to review each Downtime Notification received from an external system. For each Downtime Notification, the AMT user decides whether to:

1. Create a new AMT Event.
2. Link to an existing (Planned) AMT Event.
3. Ignore the imported notification.

Note: When importing Downtime Events with the intent to use the Downtime Import Wizard in AMT, it is necessary to provide a **Site Name** for the functionality to work correctly.

AMT also provides a means to send Downtime Events recorded in AMT to an external system.

2.4 Usage Profiles

For all AMT-ERP integrations, it is assumed that the ERP (SAP, other ERP or other data source) will be the source of the usage profile codes and descriptions, units of measure and associated annual usages. The ERP will be the master data source for usage profiles.

Master data systems may be of two types:

1. **Simple** - Solutions that hold only one annual usage at any point in time, e.g. SAP, and this annual usage applies for all future years.
2. **Complex** - Solutions that can hold different annual usages for different nominated periods into the future.

A configuration option in the adapter (see below) will control whether the master system sends simple or complex usage profiles. It is not possible to mix the type of usage profile data sent.

3 Decision Points and Configuration Options

The following AMT configuration setting will affect the behaviour of these interfaces, so the implementer will need to know these settings. When the Downtime records are imported, AMT provides the ability to create Downtime Events automatically in AMT or manually using a Wizard in AMT.

3.1 Usage

Function	Description
DaysUsageToOverwrite	Determines the number of days prior to the most recent usage reading for which usage readings can be replaced. For example, if the setting is 10, AMT will only replace readings in the last 10 days.
ProcessUsageReadingV2	Located in AssetStatisticIntegrationService.exe.config, valid values are 'True' or 'False'. When set to True, the adapter will use a new implementation for performance improvements.

3.2 Downtime

Function	Description
AutoCreateExtEvent	Determines if AMT Downtime Events are automatically created from imported Downtime records.
ImportEventMergeMinutes	Determines the tolerance (in minutes) that would merge events where the UP time of one downtime is within the specified minutes of the DOWN time of the next downtime for an Asset.
DowntimeImportMergeEvents IntoExternalEvent	Determines if the Downtime import will run a routine to merge AMT manually created Events into the imported (External) Events.

3.3 Usage Profile

Function	Description
IntegrationParam	Located in <InstalledDirectory>AssetStatisticIntegrationService.exe.config. Settings values of Simple or Complex determine processing logic for usage profiles.

4 Prerequisites

The following AMT configuration items must be completed before using these integration endpoints:

- Asset
- Unit of Measure
- Element Code
- Measurement Type
- Compartment Name
- External to AMT Asset/Site mapping

5 Integration Patterns

5.1 Supported Functions

Function	Description
ProcessUsageReading	Adds New usage readings to AMT or Updates existing usage readings in AMT, provided business rules are satisfied. The response to this function is RespondUsageReading.
ProcessUsageProfile	This function can be used to create New usage profiles or Update existing usage profiles. The response to this function is RespondUsageProfile.
GetUsageReading	This function can be used to Get usage readings recorded in AMT. The response to this function is ShowUsageReading.
ProcessMeasurement	This function is used to Add new measurement records to AMT. The function also Updates measurement records if the maintenance code and measurement time matches existing AMT records. The response to this function is RespondMeasurement.
ProcessMeasurement	This function is used to Add new measurement records to AMT. The function also Updates measurement records if the maintenance code and measurement time matches existing AMT records. The response to this function is RespondMeasurement.
GetDowntimeEvent	This function can be used to retrieve Downtime events as recorded in AMT.

5.2 Integration service address

More details regarding the general features of the integration service can be found in the Service Overview documentation. In order to use this service, the following endpoints are available for various authentication scenarios. These may not all be available, depending on the deployment configurations at a specific site.

Basic authentication	Domain, username and password provided when establishing the connection
https://<Server>:<port>/AssetStatisticIntegrationService/AssetStatisticIntegration/BasicHttpBinding_Https_Basic	
http://<Server>:<port>/AssetStatisticIntegrationService/AssetStatisticIntegration /BasicHttpBinding_Http_Basic	

Windows authentication	The Windows credentials of the current Windows account are used to establish the connection. This is the default method if no binding is specified in the URL.
https://<Server>:<port>/AssetStatisticIntegrationService/AssetStatisticIntegration/	
http://<Server>:<port>/AssetStatisticIntegrationService/AssetStatisticIntegration/	

OAUTH2 authentication	Client authentication is used to retrieve an OAUTH2 access token. The token is passed in the SOAP header when calling the AMT EPF Adapters.
https://<Server>:<port>/AssetStatisticIntegrationService/AssetStatisticIntegration/BasicHttpBinding_Https_Token	
http://<Server>:<port>/AssetStatisticIntegrationService/AssetStatisticIntegration /BasicHttpBinding_Http_Token	

<Server> is the fully qualified name of the server hosting the service.

Note that Windows authentication is being deprecated.

5.3 Usage scenarios

Synchronous integration should align with the transaction pattern as shown in Figure 1.

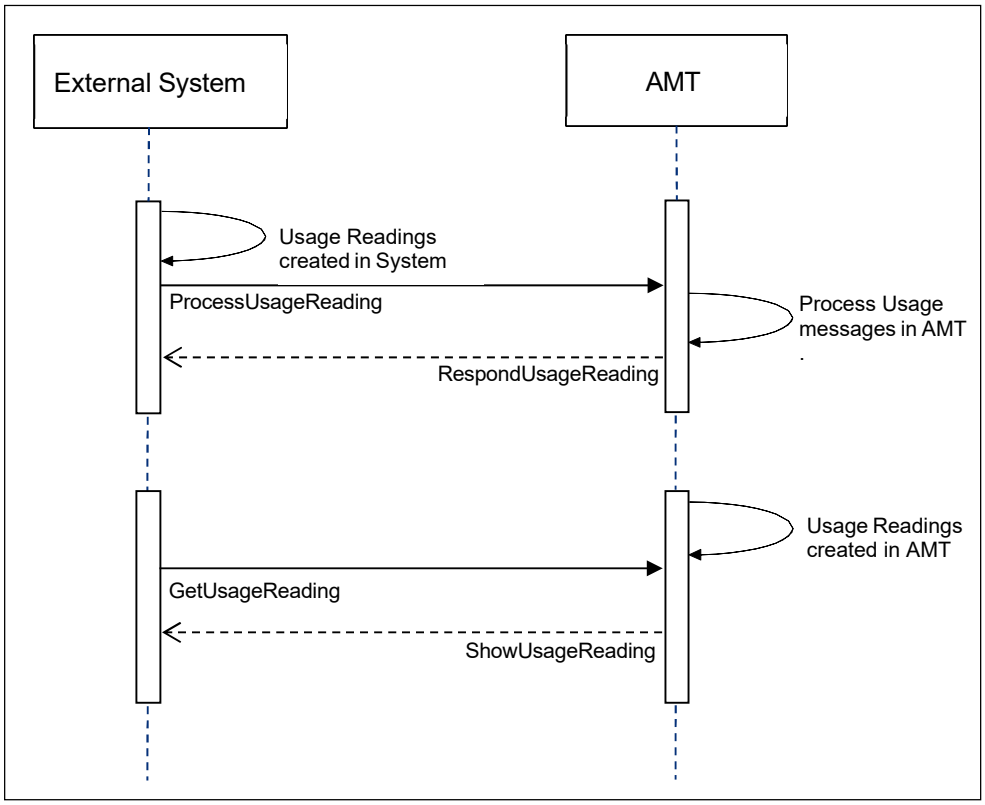


Figure 1 Synchronous Integration – Usage

Asynchronous integration using a message broker should align with the transaction pattern as shown in Figure 2.

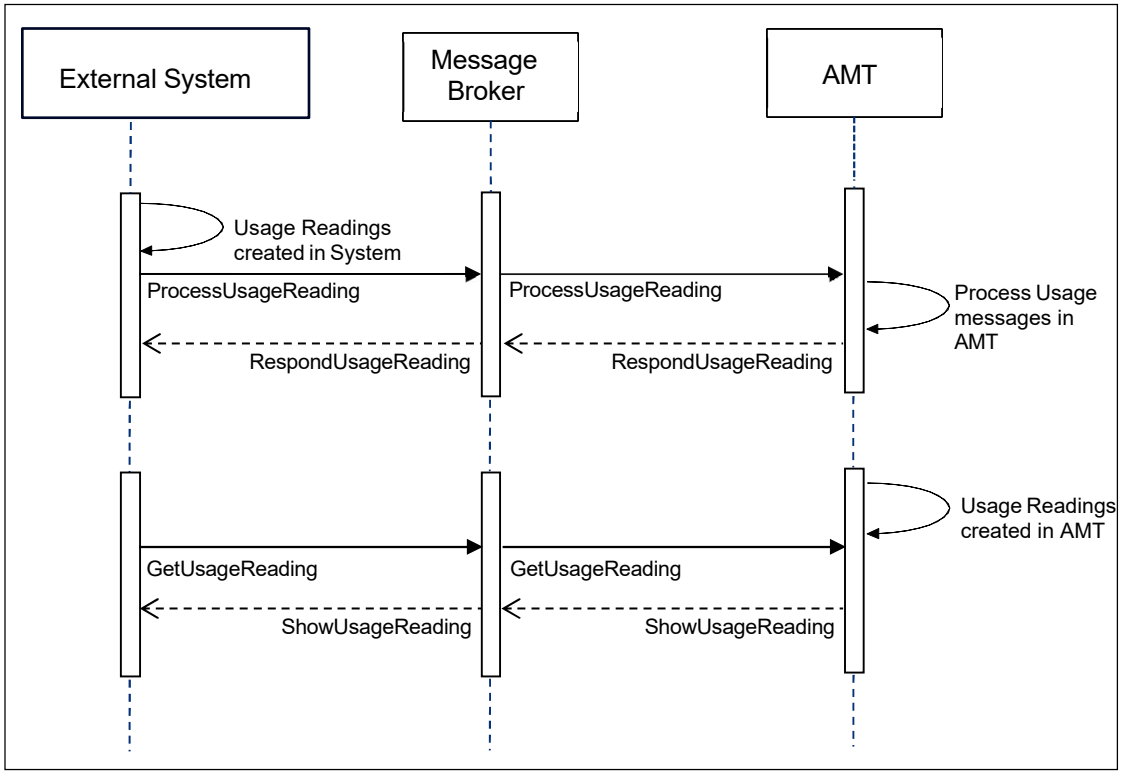


Figure 2 Asynchronous Integration – Usage

5.4 Measurement scenarios

Synchronous integration should align with the transaction pattern as shown in Figure 3.

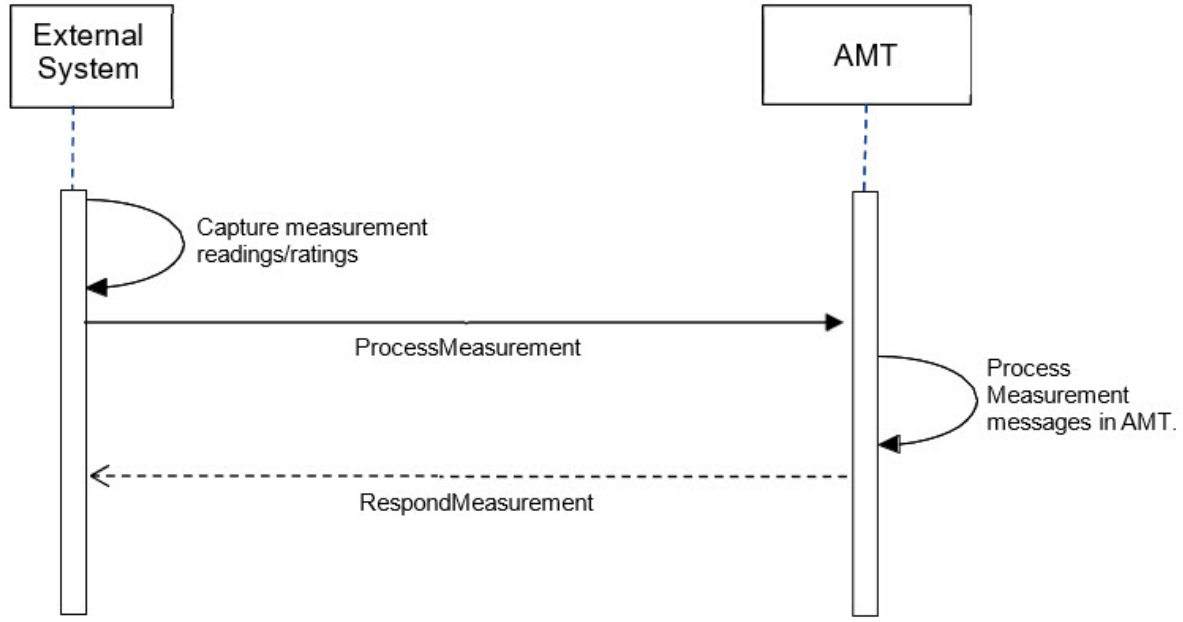


Figure 3 Synchronous Integration – Measurements

Asynchronous integration using a message broker should align with the transaction pattern as shown in Figure 4.

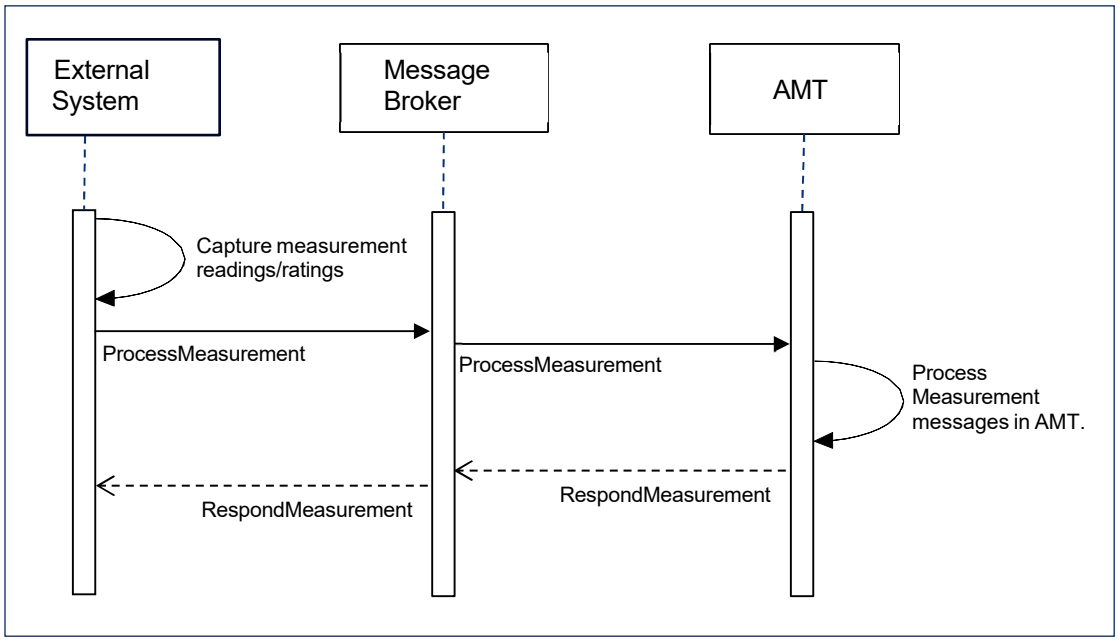


Figure 4 Asynchronous Integration – Measurements

5.5 Downtime scenarios

Synchronous integration should align with the transaction pattern as shown in Figure 5.

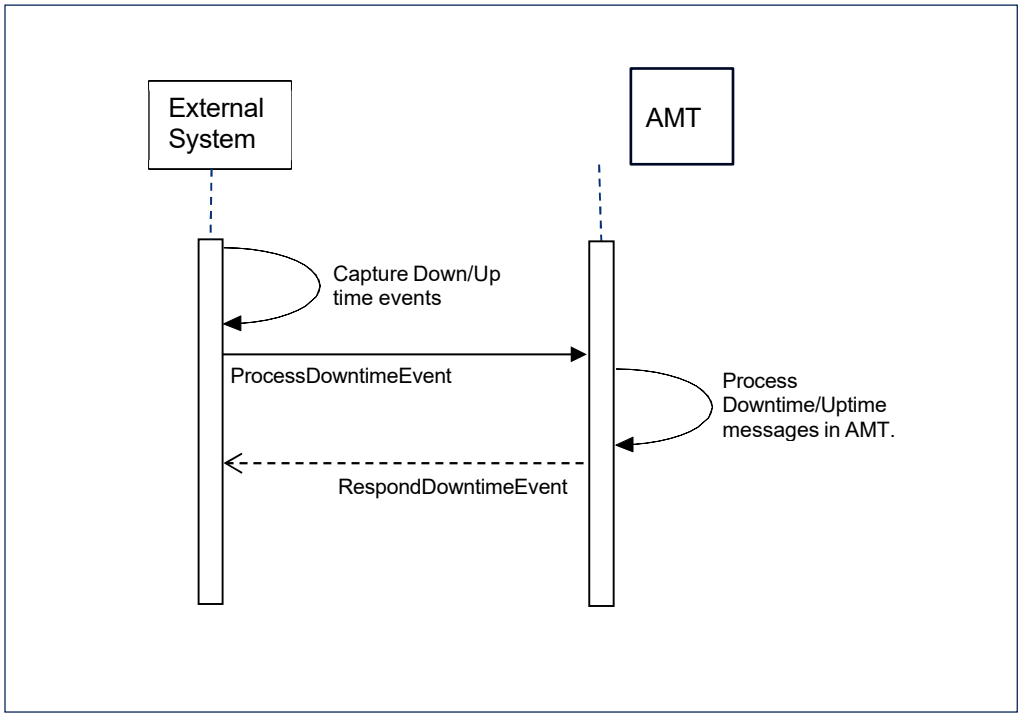


Figure 5 Synchronous Integration – Downtime Events

Asynchronous integration using a message broker should align with the transaction pattern as shown in Figure 6.

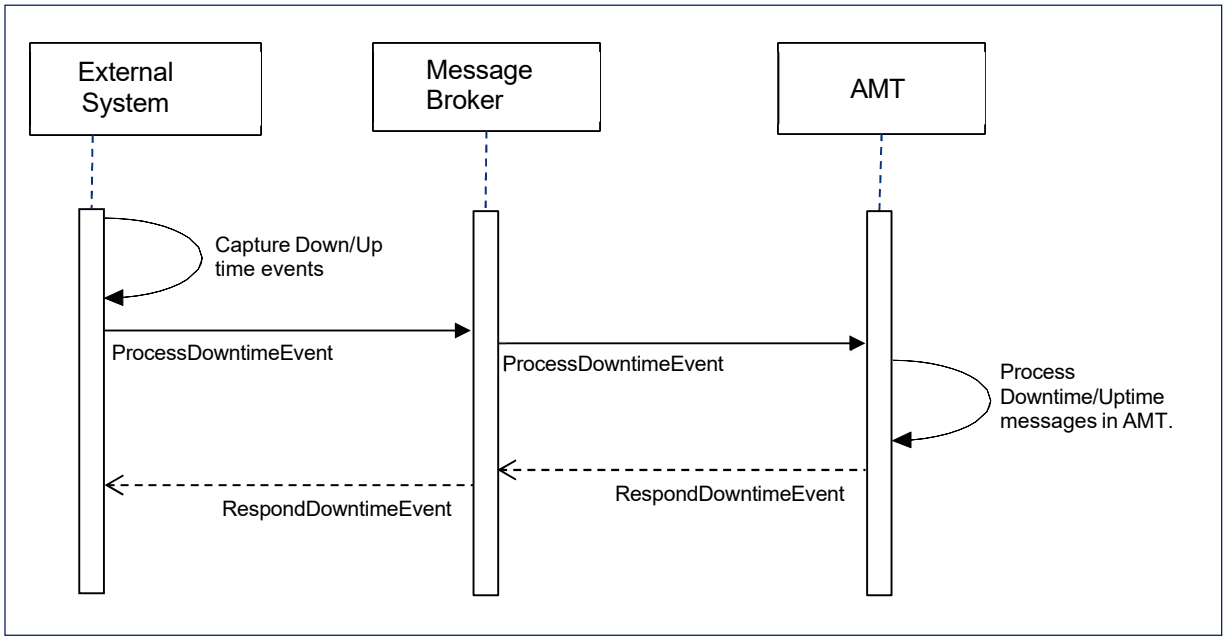


Figure 6 Asynchronous Integration – Downtime Events

6 Field Descriptions

The fields in the following tables are used for this integration. The data types for each field are described in the relevant XML schema definition.

6.1 Asset Identification

An asset is identified by:

Field Name	Description	Sample Values
ModelCode	The Asset Model Code as configured in AMT.	777D
SerialNumber	Manufacturer's Serial Number for the Asset or any other identifier for the Asset	01HK01234
RegistrationCounter	Registration Counter of the Asset, also called "Contract Enrolment Number".	Contract01

6.2 Measurement Identification

A measurement is identified by a unique combination of:

Field Name	Description	Sample Value
ElementCode	The Element whose reading/rating is measured.	Cu, Fe
UOMCode	The Unit of Measure for the reading/rating.	ppm
MeasurementTypeCode	Measurement type defines the measurement source (for example, Engine Oil, Tyre Pressure).	EO
CompartmentName	Compartment where the measurement sample was taken from.	Engine

6.3 Usage Reading

Field Name	Description	Sample Value
AssetID	The AMT Asset that the reading relates to.	
AssetID	The AMT Asset that the reading relates to.	

6.4 Meter Reading

Field Name	Description	Sample Value
MeterReadingDate	The precise date and time for which the usage reading applies in AMT local time.	2018-08-10 10:05:00
MeterReadingDateUTC	The precise date and time for which the usage reading applies in UTC time.	2018-08-10 02:05:00
MeterUOMCode	The unit of measure in which the usage reading is expressed.	HR KM
MeterReadingValue	This is the usage reading. The usage reading can either be the actual meter reading or the cumulative meter reading. They would largely be the same except for a change-out meter, where the meter reading will be less than the cumulative reading. If the actual meter reading is sent, the assumption is that the offset will be set in AMT manually.	10520
MeterTypeCode	Restricted to "C" (Cumulative) or "A" (Actual Meter). This qualifies the type of reading sent.	C

6.5 UsageReadingFilter

Field Name	Description	Sample Value
BranchDescription	Branch Description	Mining
SiteDescription	Site Description	Mobile
FleetDescription	Fleet	Haul
AssetName	Asset Name	TR50
ModelCode	Model	777D
FilterPeriod	Filter Period (from and to dates)	

6.6 FilterPeriod

Field Name	Description	Sample Value
FromDate	Filter Start Date	2017-08-01
ToDate	Filter End Date	2017-08-15

6.7 Measurement Reading

Field Name	Description	Sample Value
ModelCode	Model Code for the Asset the WO is raised for.	793C
SerialNumber	Asset Serial Number. This can be the Manufacturer's Serial Number or any other Unique Identifier for the Asset.	0A123456X
RegistrationCounter	Used to track multiple "lives" of an asset.	
MeasurementDateTime	Date and time when the measurements were taken in AMT local time.	2017-11-16 10:25
MeasurementDateTime UTC	Date and time when the measurements were taken in UTC time.	2017-11-16 02:25
AssetUsage	The meter reading for the Asset (value and UOM) when the measurements were taken.	23456
HourOnOilValue	This defines the duration (in hours) the sampled oil was in the compartment when the measurements were taken. Used for oil samplings.	10
ElementCode	The Element whose reading/rating is measured.	Cu, Fe
UOMCode	The Unit of Measure for the reading/rating.	ppm
MeasurementTypeCode	Measurement type defines the measurement source (e.g. Engine Oil, Tyre Pressure, etc).	EO
CompartmentName	The compartment from which the measurement sample was taken.	Engine
ReadingValue	Measurement Reading for the Measurement Code.	0.02
RatingValue	Measurement Rating for the Measurement Code.	N
Comment	Any comments that were noted against the recorded measurement.	Normal readings
Recommendation	Any recommendations noted against the recorded measurement.	Sample oil again in 6 weeks

6.8 Downtime

Field Name	Description	Sample Value
EventID	Unique identifier of the downtime event.	DTA123456
EventStartDateTime	The date and time that the Asset went down in AMT local time. For ProcessDowntimeEvent(), this maps to actual event start time. For showDowntimeEvent, this maps to planned event start time.	2017-11-26 18:30:27
EventStartDateTimeUTC	The date and time that the Asset went down in UTC time. For ProcessDowntimeEvent(), this maps to actual event start time. For showDowntimeEvent, this maps to planned event start time.	2017-11-26 08:30:27
EventEndDateTime	The date and time that the Asset became operational again in AMT local time. For ProcessDowntimeEvent(), this maps to actual event end time. For showDowntimeEvent, this maps to planned event end time.	2017-11-26 18:30:27
EventEndDateTimeUTC	The date and time that the Asset became operational again in UTC time. For ProcessDowntimeEvent(), this maps to actual event end time. For showDowntimeEvent, this maps to planned event end time.	2017-11-26 08:30:27
ExternalEventCode	The external system's downtime event code.	MAINT
ExternalEventCodeDescription	The external system's downtime event code description.	Maintenance
EventDescription	The description for the downtime event.	Oil seal Leak
EventNotificationDate	The reporting date and time for the downtime event in AMT local time.	2002-09-16 18:35:00
EventNotificationDateUTC	The reporting date and time for the downtime event in UTC time.	2002-09-16 10:35:00
SiteName	External site name for the asset whose downtime/uptime is reported.	BHI
AssetName	External asset name whose downtime/uptime is reported.	TR01
EventPlanCode	Defines the plan code for the event – Unplanned, Planned or Breakdown.	2
ComponentCode	Defines the AMT Component Code that the downtime is associated with.	1000

Field Name	Description	Sample Value
TaskTypeCode	Defines the AMT Task Type with which the downtime is associated.	CC
ResponsibilityCode	Defines which party is responsible for the time consumed when maintenance is performed on an asset.	Dealer
ActivityCode	Activity within a downtime allocation.	Blasting
VarianceCauseCode	The reason why a particular activity within a downtime allocation took more or less time to complete than expected.	Scheduled Blast

6.9 UsageProfile

A usage profile is uniquely identified by the usage profile name.

Field Name	Description	Sample Value
UsageProfileName	Unique identifier for the Usage Profile in the organisation.	HT793C
UsageProfileNote	Notes for the Usage Profile.	
OrganisationCode	Unique organisation code associated with the Usage Profile. The organisation code identifies where the usage profile can be used.	Will default to 1 if not provided
AssetID	Equipment associated with the usage profile.	

6.10 UsageSteps

A Usage Profile may have one or many usage steps.

Field Name	Description	Sample Value
UOM	External UOM code for the usage profile step.	H (for hours), FL (for fuel), KM (for kilometres)
ParentUOM	External UOM code for the Parent UOM for the step.	H (for hours), FL (for fuel), KM (for kilometres)
Description	Usage step description.	
StartDate	Usage step start date for the UOM.	
UsagePerCalendarYear	Usage step usage for UOM	6000

7 Error Handling

Error Condition	Description
The provided document has an unrecognised format. Various XML format errors are returned.	Check the format of the schema document against those provided. Errors can be related to well-formed XML issues or failure to validate the document against the XSD.
The provided value for <ElementCode> does not have the correct datatype.	Check datatypes of elements, especially date formats. Note that date and datetime conventions are used.
Mandatory element <ElementName> not provided.	An expected XML element has not been provided.
Mandatory attribute <AttributeName> not provided.	An expected XML attribute has not been provided.
The provided code for <ElementCode> does not have a corresponding AMT configuration item.	AMT makes extensive use of the pre-configured code list. Relevant codes must be preconfigured prior to use of the interface.
Asset does not exist for: SerialNumber: <SerialNumber> ModelCode: <ModelCode> RegistrationCounter: < RegistrationCounter > Reading rejected	The asset specified does not exist in AMT.
A more recent reading exists for asset: SerialNumber: <SerialNumber> ModelCode: <ModelCode> RegistrationCounter: < RegistrationCounter > Reading rejected	This indicates that AMT already has a reading that is more recent than the one provided.
A reading outside the permitted tolerance exists for asset: SerialNumber: <SerialNumber> ModelCode: <ModelCode> RegistrationCounter: < RegistrationCounter > Reading marked as Invalid	This indicates that the new reading is more than 500% above the projected usage in a period.
Asset has been terminated, so it cannot receive usage measurements: SerialNumber: <SerialNumber> ModelCode: <ModelCode> RegistrationCounter: < RegistrationCounter > Reading rejected	AMT cannot accept UsageReadings against terminated equipment.
The usage reading provided is less than the latest value: SerialNumber: <SerialNumber> ModelCode: <ModelCode> RegistrationCounter: < RegistrationCounter > Reading marked as Invalid	Usage readings can only increase value.

Error Condition	Description
The provided Measurement Code does not exist ElementCode: <ElementCode> UOMCode: < UOMCode MeasurementTypeCode:<MeasurementTypeCode > CompartmentName: < CompartmentName >	The Measurement Code (Measurement, Element, Compartment combination) does not exist in AMT for the readings/ratings to be applied.

8 Integration functions

8.1 ProcessUsageReading

This interface is used to add new usages to AMT or update existing usages in AMT.

8.1.1 Business Rules

8.1.1.1 General Business Rules

AMT does some business rule validations on the set of readings being imported.

- AMT does not update readings in the past.
- Actual readings imported must be greater than or equal to the previous readings recorded for each asset.
- For terminated Equipment, readings imported cannot be greater than End Usage.
- The reading must be within the "UsageTolerance" set in AMT. The tolerance is defined as a percentage value and by default is set to **500**. The tolerance is configurable**.**

8.1.2 Field Level Rules

Field Name	Business Rules	Optional/ Required
ModelCode	The value provided must match a configured Model code within AMT. Forms the key along with SerialNumber and RegistrationCounter.	R
SerialNumber	Forms the key along with ModelCode and RegistrationCounter.	R
RegistrationCounter	Forms the key along with ModelCode and SerialNumber. If the Registration Counter provided does not match that of an Asset in AMT (in conjunction with ModelCode and SerialNumber), the active Equipment is selected. If no active Asset exists, an error is returned.	O
MeterReadingDate	This is a key field and, together with AssetID and UOMCode, uniquely identifies a Usage record within AMT. Can be omitted if MeterReadingDateUTC is provided.	R
MeterReadingDateUTC	As an alternative to the MeterReadingDate, a UTC value can be provided. This will be converted to local time by the system.	
MeterUOMCode	This is a key field and, together with AssetID and UsageReadingDate, uniquely identifies a Usage record within AMT.	R
MeterReadingValue	This is the Meter Reading. Depending on the MeterReadingTypeCode, this could be the Cumulative reading or the Actual reading. This value must be greater than the existing reading stored in AMT.	R
MeterReadingTypeCode	Only the following values are accepted: • C – Cumulative Readings • A – Actual Readings	R

8.1.3 Processing Rules

1. One or more Asset Usage readings can be updated at the same time.
2. A usage reading cannot be imported for terminated equipment.
3. All usages for an Asset must be successful; otherwise, the Asset Usage update returns an error.
4. Usage readings earlier than those already captured cannot be imported. The AMT configuration described below should be used to correct data retrospectively.
5. Consider the AMT Configuration Item *DaysUsageToOverwrite*.
   a. The most recent usage record for an asset in AMT is used as the basis for the calculation.
   b. The *DaysUsageToOverwrite* value is used to calculate the earliest date that records are considered for deletion (a – *DaysUsageToOverwrite)*.
   c. The earliest date of usage readings in the import (for an asset) is compared to the date calculated above.
   d. All records after the later of the two dates are deleted.
   e. Records after this date are imported from the import message.
   f. If *DaysUsageToOverwrite* is 0, only readings later than the last stored reading will be imported**.**
6. The following status is returned for each usage reading. These are all valid return codes and will result in Successful processing of the asset usage import.

Return Code	Description	AMT Action
Valid	The Usage has been successfully updated.	Added to AMT
Invalid	The new reading is more than 500% above the projected usage in a period, or the new reading is less than the previous reading.	Added to AMT as invalid reading
Rejected	A more recent reading exists.	Rejected

7. RespondUsageReading is returned upon completion of processing.

8.1.4 Data Structure

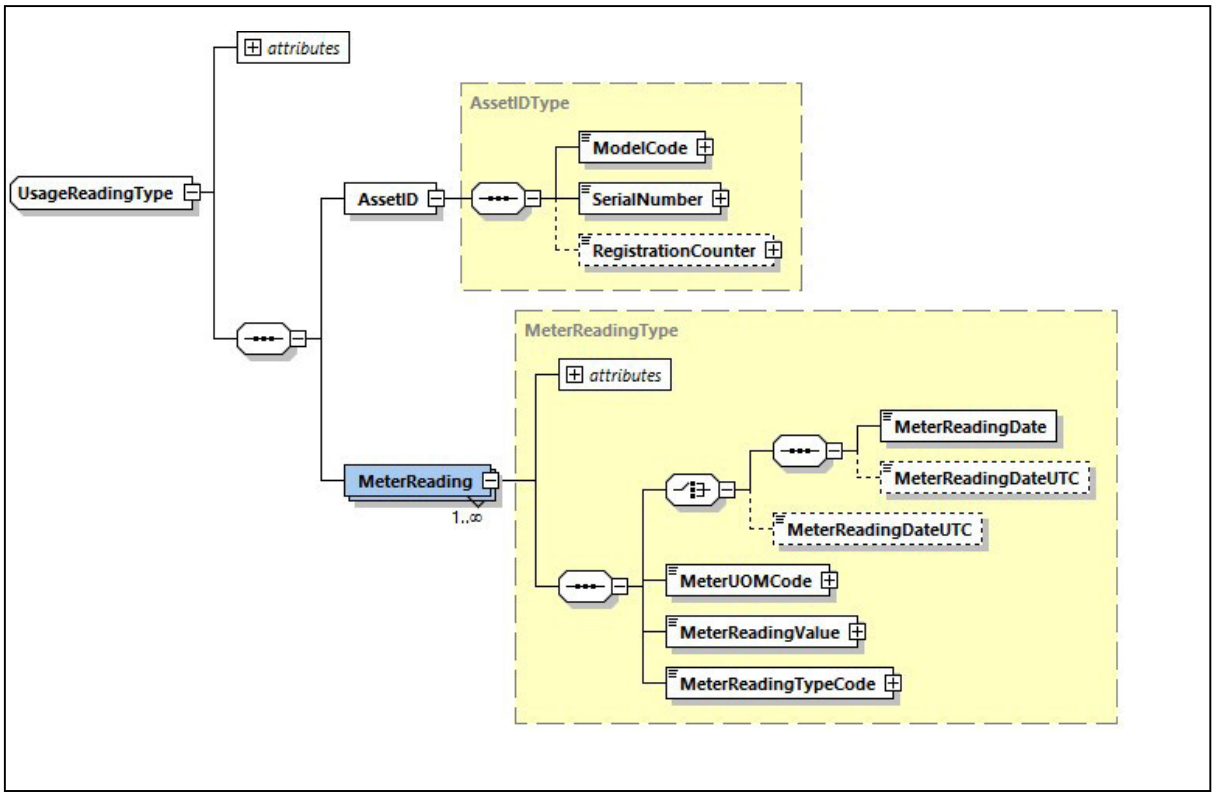


Figure 7 ProcessUsageReading

8.2 GetUsageReading

This interface is used to get usage readings recorded in AMT. Filter criteria can be passed in the interface to filter what usages to get from AMT.

8.2.1 Business Rules

Field Name	Business Rules	Optional/ Required
BranchName	The values provided must match configured Branch codes within AMT.	O
SiteName	The values provided must match configured Site codes within AMT.	O
FleetName	The values provided must match configured Fleet codes within AMT.	O
AssetName	One or more Assets.	O
ModelCode	The values provided must match configured Model codes within AMT.	O
FromDate	Filter Start Date in AMT local time. If not provided, get all usage readings up to the ToDate (if provided).	O

8.2.2 Processing Rules

1. Query AMT for valid Usage readings based on the filter condition provided.
2. All items within a List element are considered as a logical OR.
3. Each List is considered as a logical AND.
4. The use of a non-existent code will result in an error condition ShowUsageReading is returned upon completion of the process.

8.2.3 Data Structure

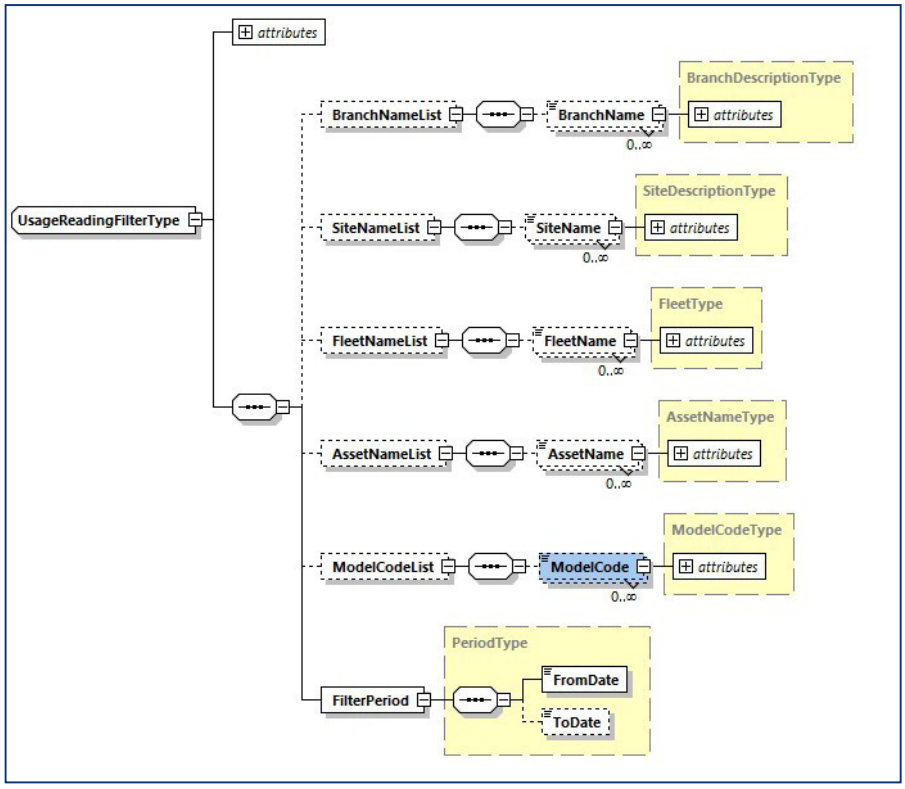


Figure 8 UsageReadingFilter

8.3 ProcessMeasurement

This interface is used to add new measurement readings or ratings to AMT. This interface can also be used to update measurement readings or ratings provided necessary business rules are adhered to.

8.3.1 Business Rules

Field Name	Business Rules	Optional/ Required
ModelCode	This field is always used to uniquely identify an asset within AMT. This field is used together with ModelCode and RegistrationCounter to uniquely identify the part within AMT.	R
SerialNumber	The value provided should match a configured Model code within AMT. This field is used together with Serial Number and RegistrationCounter to uniquely identify the asset within AMT.	R
RegistrationCounter	This field is used together with Serial Number and ModelCode to uniquely identify the asset within AMT. If not provided, the currently active registration is used.	O
MeasurementDateTime	Can be omitted if MeasurementDateUTC is provided.	R
ElementCode	This field is used together with MeasurementTypeCode, CompartmentName and UOMCode to identify the MeasurementCode.	R
MeasurementTypeCode	This field is used together with ElementCode, CompartmentName and UOMCode to identify the MeasurementCode.	R
CompartmentName	This field is used together with MeasurementTypeCode, ElementCode and UOMCode to identify the MeasurementCode.	R
UOMCode	This field is used together with MeasurementTypeCode, CompartmentName and ElementCode to identify the MeasurementCode. If no value is provided, the Asset default is used.	O
ReadingValue	Reading recorded for a measurement. Where only a Rating is recorded and sent for a measurement, the external system also needs to send a reading, even if it is a Dummy reading (usually sent as 0).	R

8.3.2 Processing Rules:

1. For each asset determine the validity of the records:
   a) "Valid" if it passes all business rules stated in the above table.
   b) "Invalid" if there are no translation rules.
   c) "Rejected" if the readings fail business rules above or are in the future.
2. Add all "valid" records to AMT.
3. Add all "invalid" records to AMT (marked as invalid).
4. Ignore all "rejected" records.
5. **RespondMeasurement** is returned upon completion of processing.

8.3.3 Data Structure

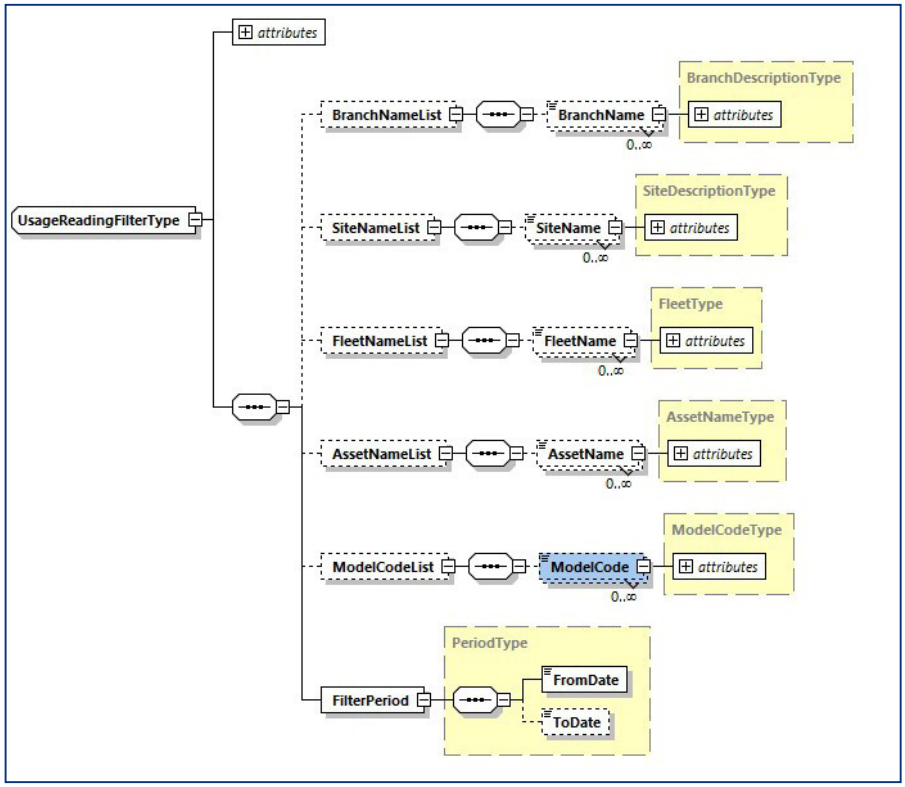


Figure 9 ProcessMeasurement

8.4 ProcessDowntimeEvent

This interface is used to add new downtime events into AMT. The interface is also used to update downtime events with uptime information based on the EventID key field.

8.4.1 Business Rules

Field Name	Business Rules	Optional/ Required
EventID	Value provided will be used to add or update an event. Events will be updated if a match on the EventID is found.	R
EventStartDateTime	Can be omitted if EventStartDateTimeUTC is provided. If provided, this value will be used to update AMT.	R
EventStartDateTimeUTC	Can be omitted if EventStartDateTime is provided. Will be ignored if EventStartDateTime is provided.	R
EventEndDateTime	If provided, this value will be used to update AMT.	O
EventEndDateTimeUTC	Will be ignored if EventEndDateTime is provided.	O
EventDescription	The description for the downtime event. The Event created will inherit this description.	R
EventNotificationDate	If provided, this value will be used to update AMT.	O
SiteName	The value provided should match a configured Site Name within AMT. **Note:** When importing Downtime Events with the intent to use the Downtime Import Wizard in AMT, it is necessary to provide a Site Name for the functionality to work correctly.	O
AssetName	External Asset Name must be mapped to an AMT Asset using the 'Configure External Downtime Equipment' function in AMT.	R
EventPlanCode	The following values are accepted: 0 – Unplanned 1 – Planned 2 - Breakdown If no value is provided or any other value is provided, a default value of "2" will be used.	O
ComponentCode	The value provided should match a configured Component Code within AMT. If no value is provided, the default AMT Component code will be used.	O
TaskTypeCode	The value provided should match a configured Task Type Code within AMT. If no value is provided, the default AMT Task Type code will be used.	O
ResponsibilityCode	The value provided should match a configured Responsibility Code within AMT. If no value is provided, the default AMT Responsibility Code will be used.	O

Field Name	Business Rules	Optional/ Required
ActivityCode	The value provided should match a configured Activity Code within AMT If no value is provided, the default AMT Activity code will be used.	O
VarianceCauseCode	If no value is provided, Variance is blank. A warning is returned. If Variance provided is not in AMT, Variance is blank. A warning is returned. If Variance does not belong to Activity provided, Variance is blank. A warning is returned.	O

8.4.2 Processing Rules

1. One or more Downtime notifications for Assets can be imported at the same time.
2. AMT will merge multiple downtime notifications for the same asset that have exactly continuous start and end date/times. For example, the following three downtime notifications for the same asset:

Event ID	Down Time	Up Time
1	27/11/2017 10:30	28/11/2017 06:00
2	28/11/2017 06:00	29/11/2017 06:00
3	29/11/2017 06:00	29/11/2017 14:30

   will be merged by AMT into *one* downtime event, as follows:

Event ID	Down Time	Up Time
1	27/11/2017 10:30	29/11/2017 14:30

3. When AMT is configured to create AMT events automatically, AMT will merge multiple downtime notifications for the same asset if the downtime and uptime are within the tolerance (in minutes) defined by the configuration variable "ImportEventMergeMinutes".
4. For the same event, identified by the EventID, if multiple notifications come with multiple Responsibilities, multiple DTA records are created within the AMT event.
5. Only one "In Progress" event can exist for an asset.
6. Up time for an event cannot be smaller than the down time for the event.
7. For Completed downtime events (that is, uptime is provided), all work orders within the event are to be set to "Complete". For In-Progress downtime events, all open work orders within the event are set to be "In Progress". Status for work orders that are "Complete" is not altered. Abandoned and deferred work order status does not change either.
8. **RespondDowntimeEvent** is returned upon completion of processing.

8.4.3 Data Structure

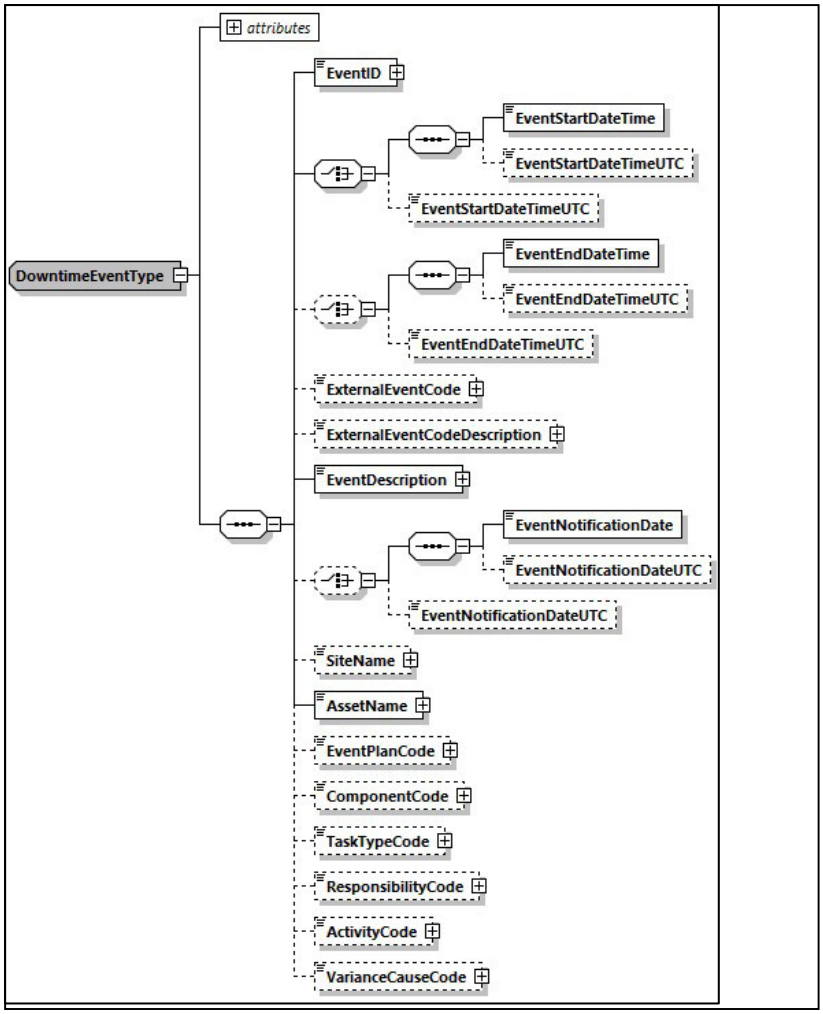


Figure 10: ProcessDowntimeEvent

8.5 GetDowntimeEvent

This interface is used to get Downtime Events recorded in AMT. Filter criteria can be passed in the interface to filter which Events to retrieve from AMT.

8.5.1 Business Rules

Field Name	Business Rules	Optional/ Required
BranchName	The values provided must match configured Branch codes within AMT.	O
SiteName	The values provided must match configured Site codes within AMT.	O
FleetName	The values provided must match configured Fleet codes within AMT.	O
ModelCode	The values provided must match configured Model codes within AMT.	O
TaskTypeCode	The values provided must match configured Task Type codes within AMT.	O
WorkLocationCode	The values provided must match configured Work Location codes within AMT.	O
WorkGroupCode	The values provided must match configured Work Group codes within AMT.	O
PriorityCode	The values provided must match configured Priority codes within AMT.	O
PlanningStatusCode	The values provided must match configured Planning Status codes within AMT.	O
ResponsibilityCode	The values provided must match configured Responsibility codes within AMT	O
FromDate		R
ToDate	If not provided, all downtime events after the FromDate will be returned	O

8.5.2 Processing Rules

1. Query AMT for Downtime Events based on the filter condition provided.
2. All items within a List element are considered as a logical OR.
3. Each List is considered as a logical AND
4. The use of a non-existent code will result in an error condition.
5. ShowDowntimeEvent is returned upon completion of the process.

8.5.3 Data Structure

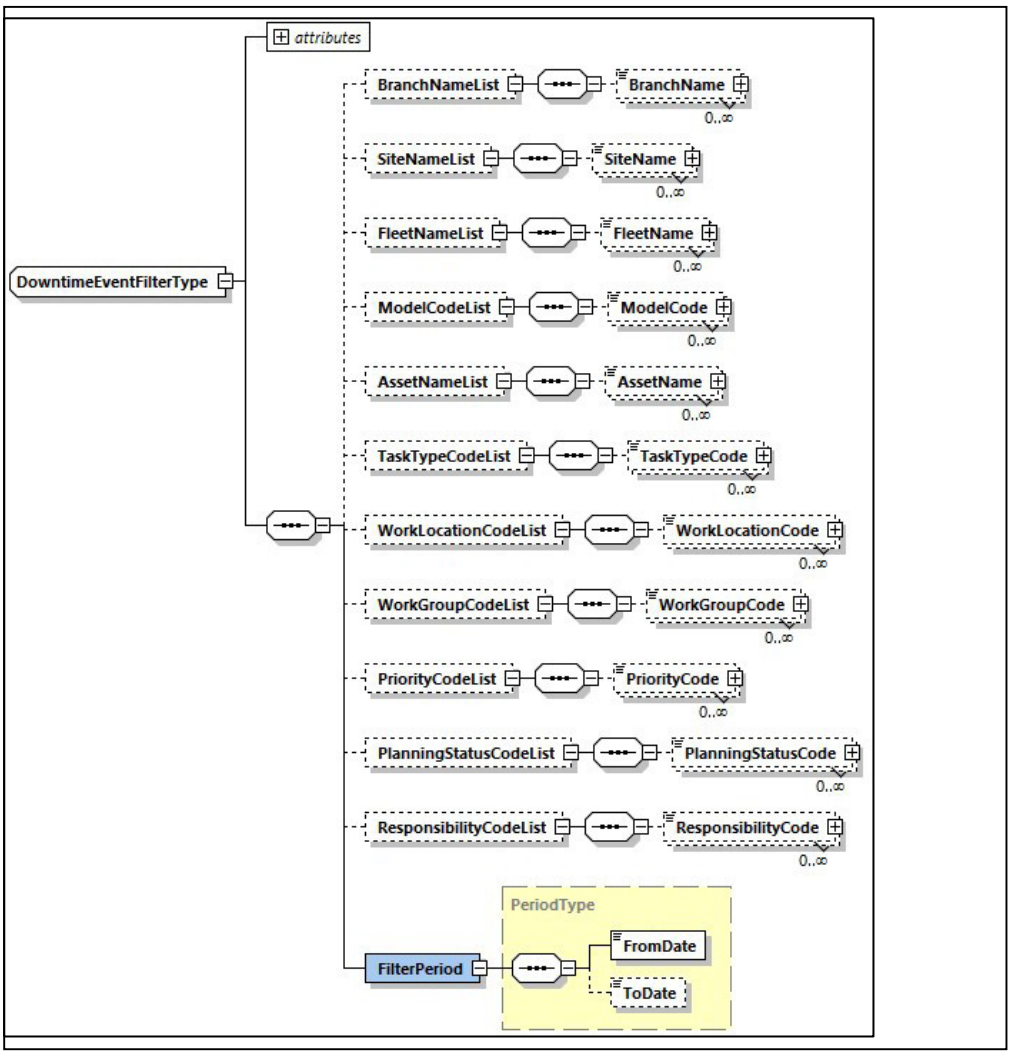


Figure 11: DowntimeEventFilter

9 ProcessUsageProfile

This interface is used to add new usage profiles to AMT or update existing usage profiles in AMT.

9.1.1 Business Rules – Usage Profile

9.1.1.1 Field Level Rules

Field Name	Business Rules	Optional/ Required
UsageProfileName	If the usage profile exists for the nominated organisation, the existing details will be updated. If the usage profile does not exist for the nominated organisation, a new profile will be created.	R
OrgansiationCode	Must be a valid AMT organisation code.	R
ModelCode	The value provided must match a configured Model code within AMT. Forms the Asset ID key along with SerialNumber and RegistrationCounter.	O
SerialNumber	Forms the Asset ID key along with ModelCode and RegistrationCounter.	O
RegistrationCounter	Forms the Asset ID key along with ModelCode and SerialNumber.	O

9.1.2 Business Rules – Usage Steps

9.1.2.1 Field Level Rules

Field Name	Business Rules	Optional/ Required
UOM	Must be a valid AMT External UOM code.	R
ParentUOM	Must be a valid AMT External UOM code.	O
StartDate	Must be a valid date. Dates earlier than the date set in the variable UsageProfileMinDate in AMT_TYPED_VARIABLE will be set to the value of UsageProfileMinDate. Cannot be later than the date set in the variable UsageProfileMaxDate.	R
UsagePerCalendarYear	Must be a valid number.	R

9.1.3 Processing Rules (IntegrationParam=Simple)

1. One or more Usage Profiles can be updated at the same time.
2. A usage profile can have one or more usage steps.
3. Usage Profile name, UOM and Start Date identify whether an existing step will be updated.
4. If a new step is being created, if an existing step has an end date overlapping the new step's start date, the existing step will be updated to end at the new step's start date.
5. If a new step is being created, the new step end date will be UsageProfileMaxDate (AMT_TYPED_VARIABLE). Any existing steps with a start date after the new step start date will be deleted. (i.e. new steps are always created as the last step and end at the maximum date).
6. If the message start date is less than the date of the integration run, the date of the integration run will be used as the start date instead of the date in the message.
7. RespondUsageProfile is returned upon completion of processing.

9.1.4 Processing Rules (IntegrationParam=Complex)

1. One or more Usage Profiles can be updated at the same time.
2. A usage profile can have one or more usage steps.
3. Usage Profile name, UOM and Start date identify whether an existing step will be updated.
4. If a new step is being created, if an existing step with an end date overlapping the new step start date is created, the existing step will be updated to end at the new step's start date.
5. If a new step is being created, if an existing step with a start date later than the new step's will end at the start date of the later step (i.e., the new step is inserted in between existing steps).
6. If a new step is being created, and there is no step with a later start date than the new step, the end date for the new step will be UsageProfileMaxDate (AMT_TYPED_VARIABLE).
7. If the message start date is less than the date of the integration run, the date of the integration run will be used as the start date instead of the date in the message.
8. **RespondUsageProfile** is returned upon completion of processing.

9.1.5 Data Structure

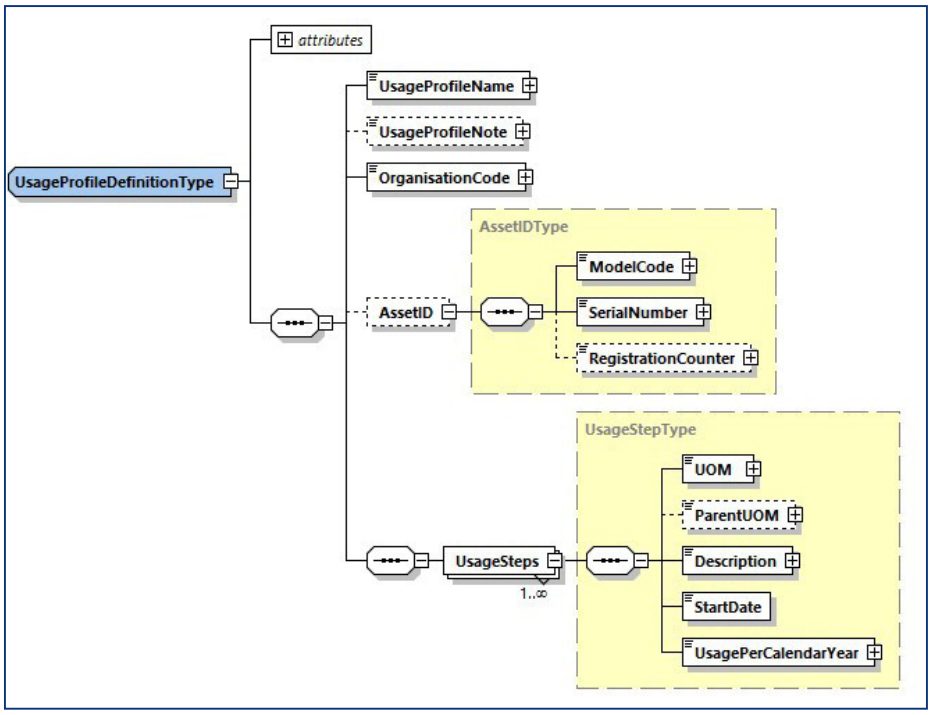


Figure 12: ProcessUsageProfile

10 Response formats

10.1 RespondUsageReading

This payload is returned in response to the following interfaces:

- ProcessUsageReading

10.1.1 Data Structure

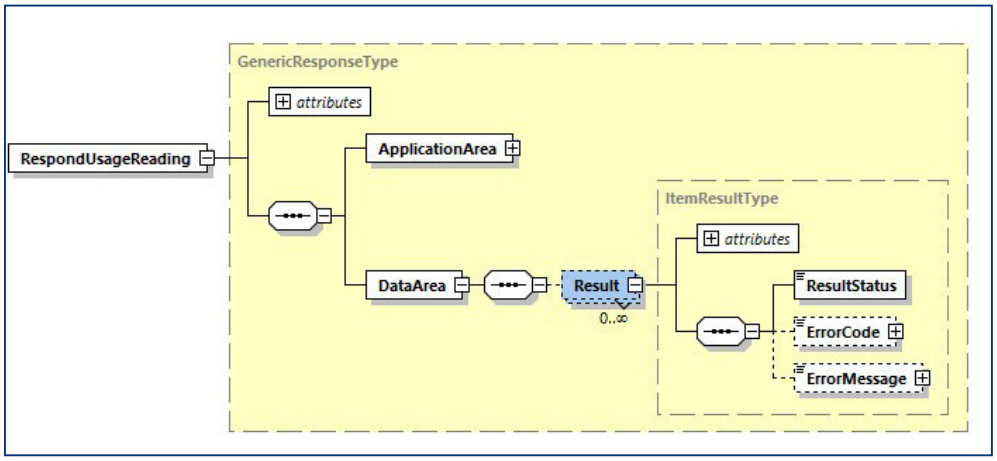


Figure 13: RespondUsageReading

10.2 ShowUsageReading

This payload is returned in response to the following interfaces:

- UsageReading

10.2.1 Data Structure

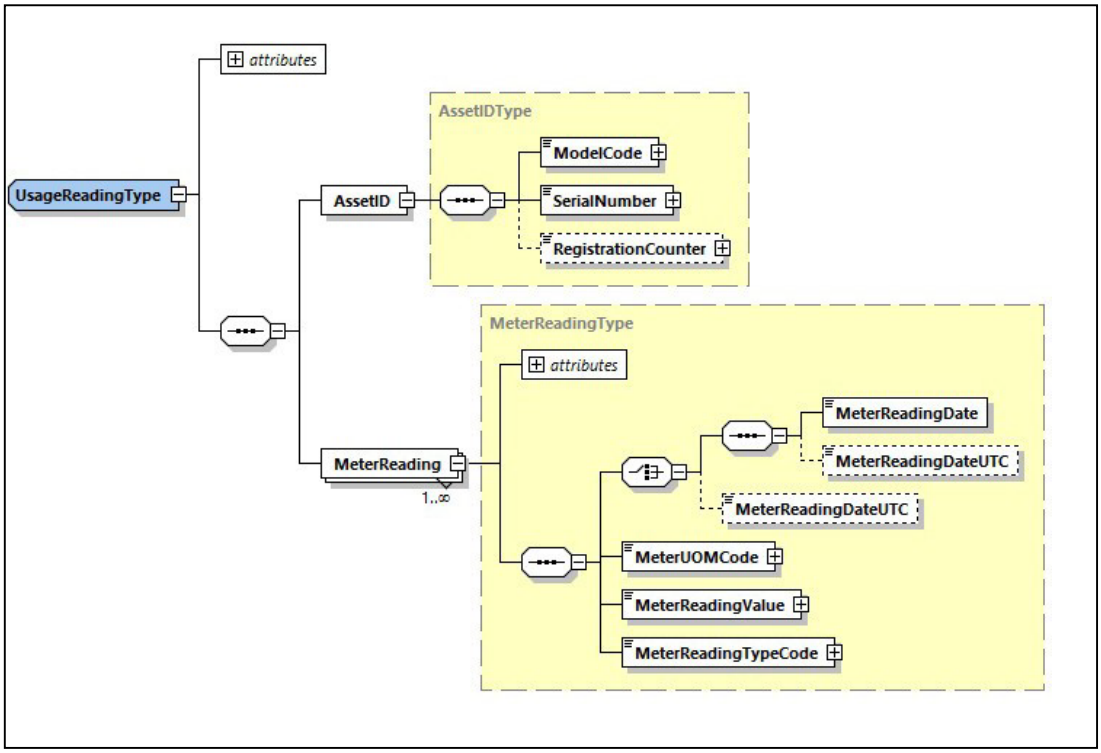


Figure 14: ShowUsageReading

10.3 RespondMeasurement

This payload is returned in response to the following interfaces:

- ProcessMeasurement

10.3.1 Data Structure

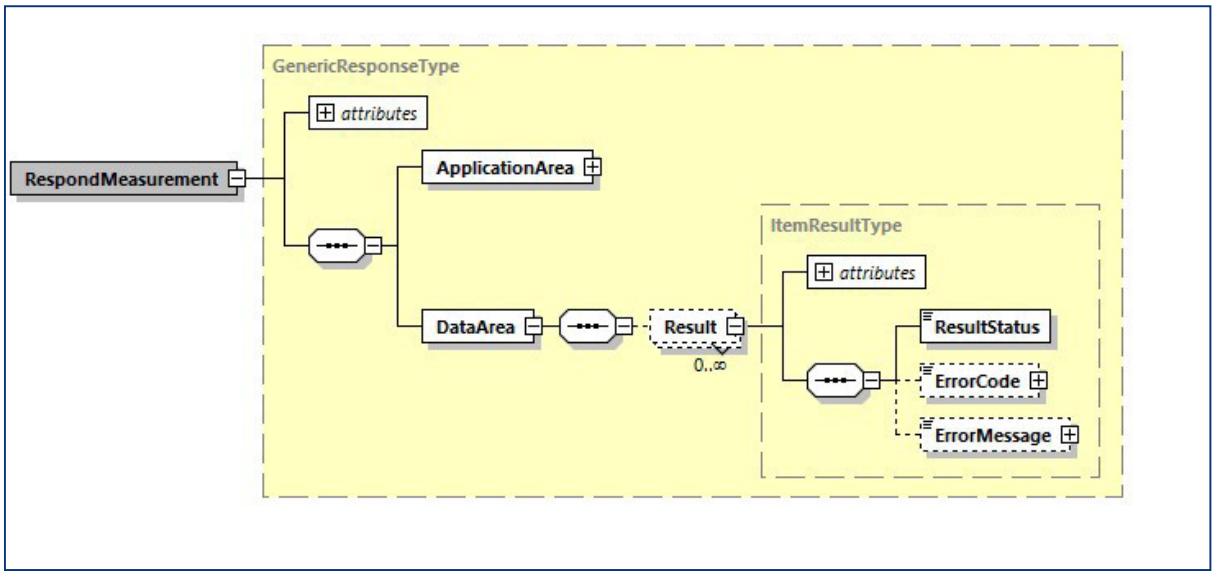


Figure 15: RespondMeasurement

10.4 RespondDowntimeEvent

This payload is returned in response to the following interfaces:

- ProcessDowntimeEvent

10.4.1 Data Structure

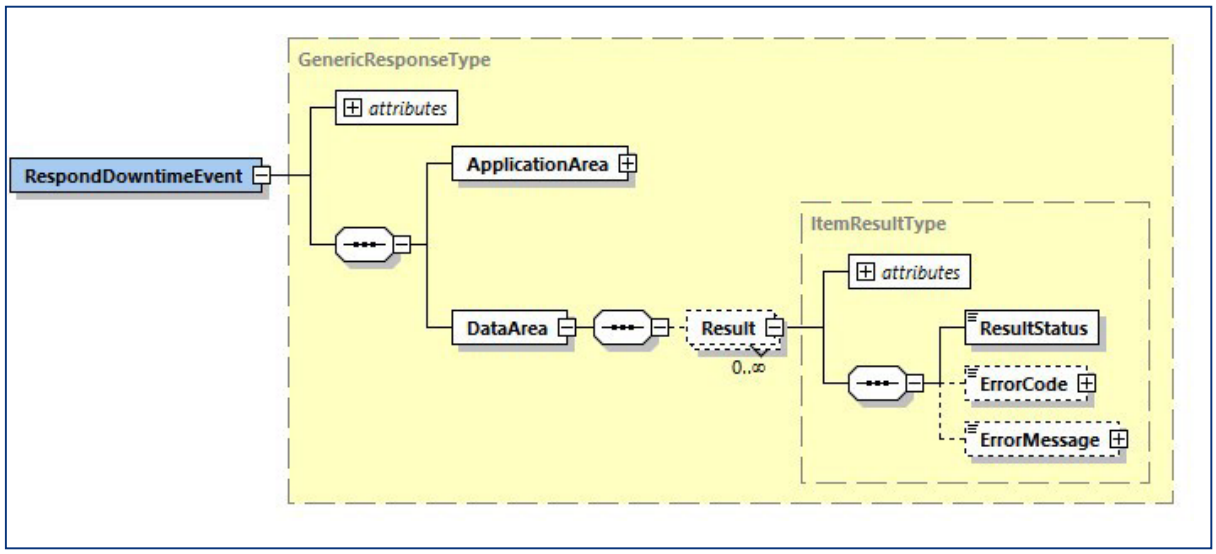


Figure 16: RespondDowntimeEvent

10.5 ShowDowntimeEvent

This payload is returned in response to the following interfaces:

- GetDowntimeEvent

10.5.1 Data Structure

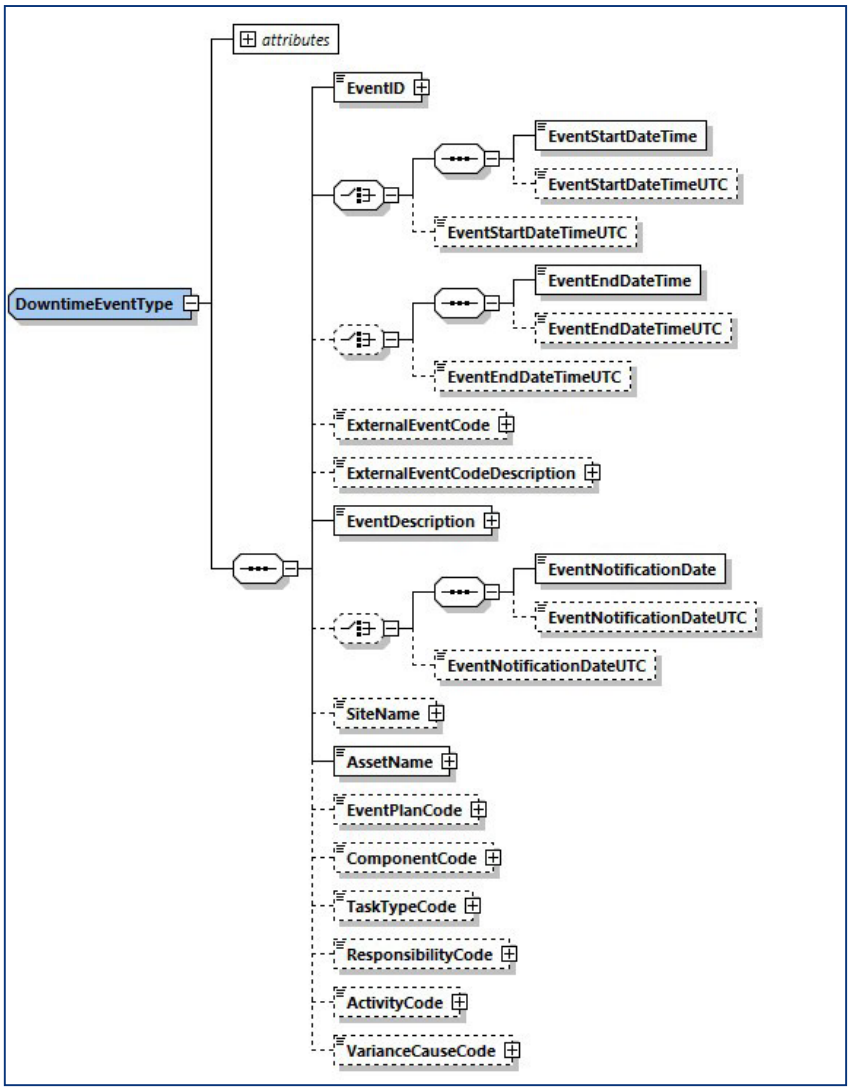


Figure 17: ShowDowntimeEvent

10.6 RespondUsageProfile

This payload is returned in response to the following interfaces:

- ProcessUsageProfile

10.6.1 Data Structure

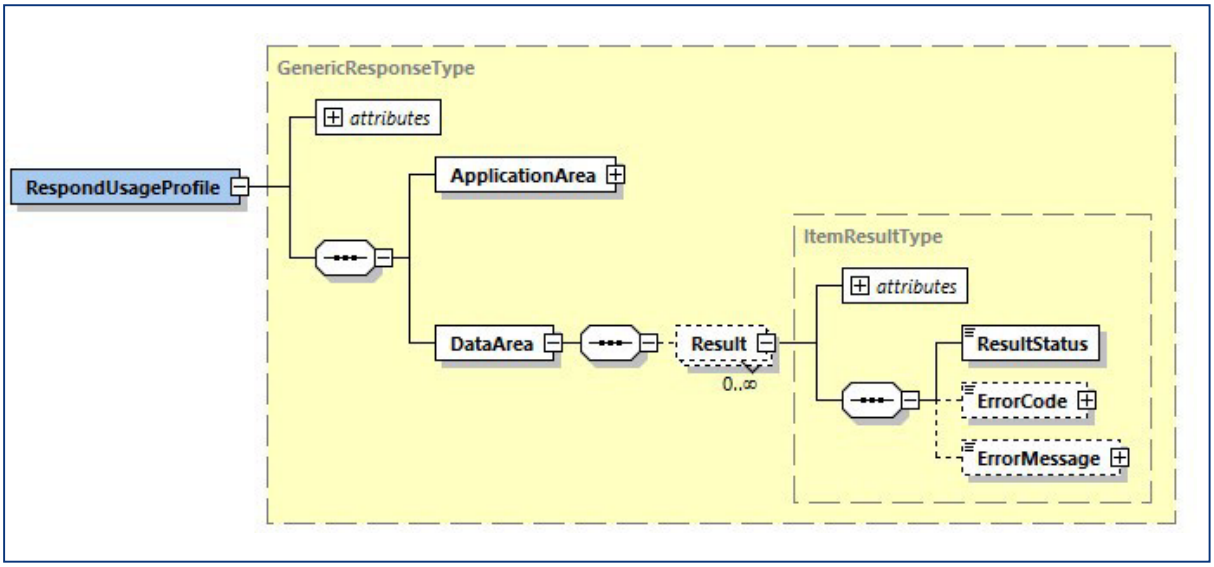


Figure 18: RespondUsageProfile